

Where the network binds, dispatching becomes road scheduling

Deadlock-free task assignment and passage sequencing: pushing up the lower edge of the capacity interval (research plan)

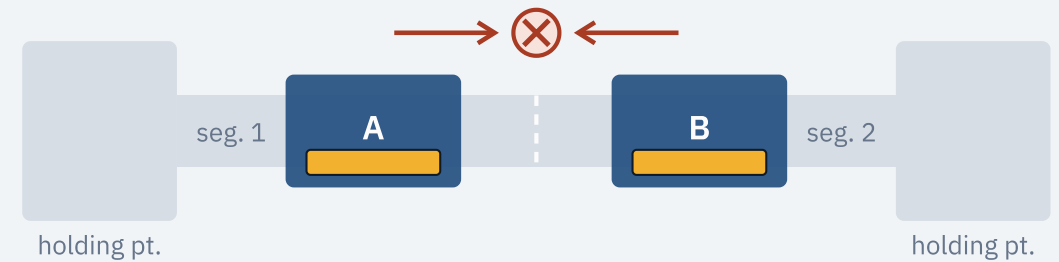
Deadlock-free task assignment and traffic orchestration for agents whose footprint grows when loaded (working title)

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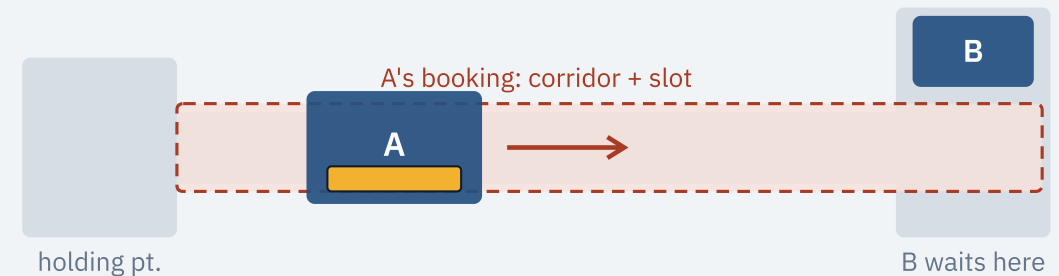
Grounded in the chapter 3 flag experiment (thesis-core 83c16bf, reviewed) and research manifesto v1

Naive segment request: head-on in one lane



A holds 1, waits for 2; B holds 2, waits for 1

T5: reserve up to the next safe holding point



The slot is booked before release: no hold-and-wait cycle

This study in one minute

1

The question

Chapter 3 shows that capacity is an interval, [best deadlock-free rule, T1']. The traffic rule sets the lower edge: on Okpo, whole-route reservation plateaus at 36% of T1'.

2

Why new rules

Naive segment request deadlocks 1,911 times a day on Okpo at $K = 100$. “Teleport” clearing takes about 30% of vehicle-hours off the network, so it is only a reference rule.

3

The approach

A safety shell (T5, safe release) rules out deadlock; inside it, task assignment and passage order are searched together, with CP-SAT ground truth on small instances and neighbourhood search at scale.

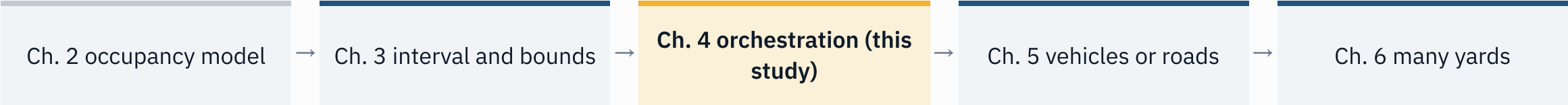
4

Outputs

A lower edge pushed towards field per-vehicle productivity; an answer to H5, what safety costs; a proof of T5; a chapter 4 journal paper.

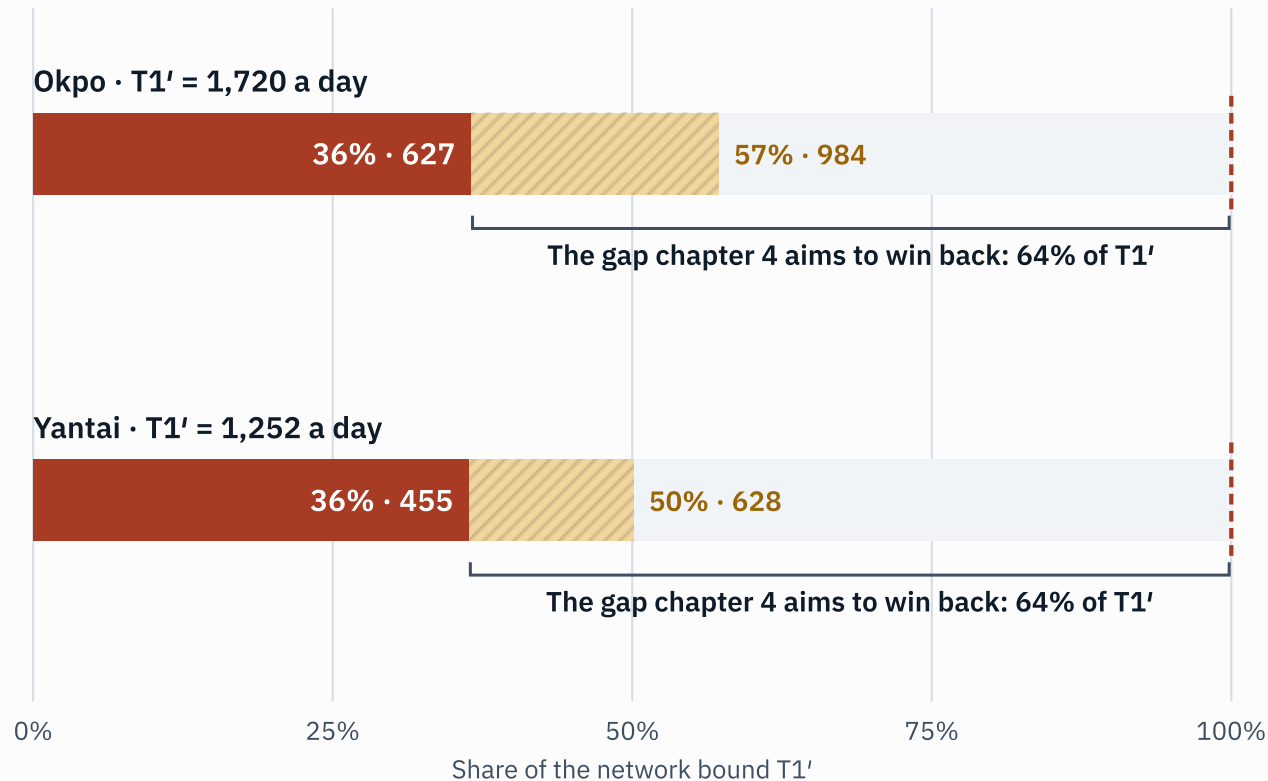
Where the network binds, how much of the bound is achieved is decided by traffic orchestration. (C3)

Its place in the thesis: chapter 3 gives the capacity interval, chapter 4 pushes its lower edge up



ITEM	MANIFESTO V1 WORDING	HOW CHAPTER 4 ANSWERS
C1 interval	Achievable capacity lies in the interval [best deadlock-free rule, network bound]	The lower edge is the rule chapter 4 delivers: first safe release, then optimised orchestration
C3 dispatching becomes road scheduling	Where the network binds, how much of the bound is achieved is decided by traffic orchestration: release order, waiting points, reservation mode, block granularity. Naive dispatching deadlocks	All four levers become decision variables; deadlocks are counted under no-teleport rules
C4 boundary	When the plateau lies far below the bound, change the orchestration first. This is the boundary between C3 and C4	Okpo's main scenario has plateau / $T1' \approx 36\%$: orchestration comes first
H5 safe orchestration	Naive dispatching deadlocks or nearly deadlocks at peaks; a provably safe policy loses only a little throughput	Both criteria and the reading: page 9
T5 theorem	Safe orchestration: reserving to the next safe holding point never deadlocks, with a loss bound relative to the upper bound	Manifesto status “to do (chapter 4)”; proof route on page 8

The traffic rule sets the lower edge of the capacity interval: on both yards the reservation plateau is only about 36% of T1'



36% / 36%

Whole-route reservation plateau ÷ T1' (Okpo / Yantai): today's deadlock-free lower edge

57% / 50%

Segment request (reference rule, teleport clearing) ÷ T1': it relies on teleporting, so it cannot be the lower edge

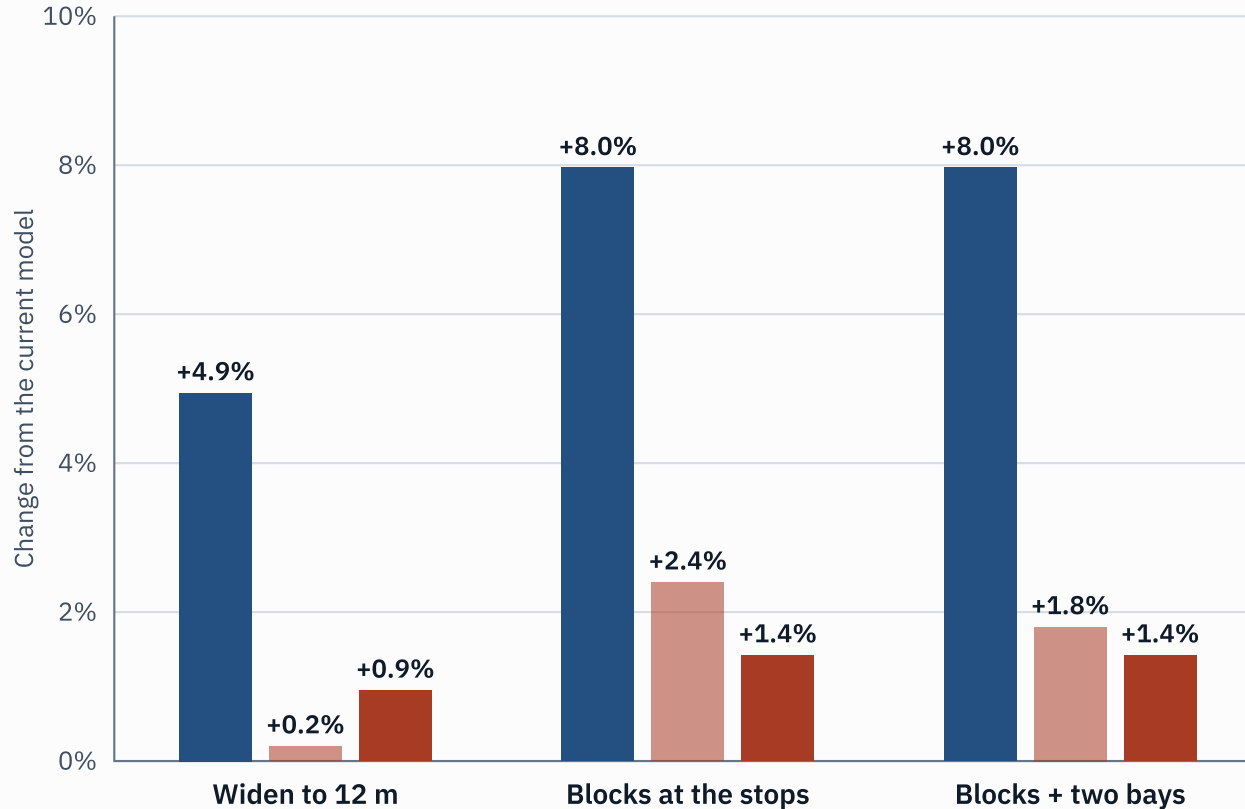
627 → 1,720

Okpo's interval (a day). Chapter 4 pushes the lower edge up; chapter 3's tighter bounds (T23) press the upper edge down

- Whole-route reservation: deadlock-free, today's lower edge
- Segment request (reference rule, teleport clearing) [] T1' network bound

Sources: thesis-core 83c16bf. Okpo results/a0ab16fceb94 and 49d2383872e7 (MY-B task mix, map rev 1824, network identical to rev 1825); Yantai results/b82aa19b22af (main scenario, map rev 395, network identical to rev 396). Conditions: saturated demand, 16-h day, one resource per road (whole segment). Plateau = mean throughput over K = 100–150 (10 seeds); T1' = mean bound over 10 seeds. Free flow ignores occupancy, exceeds T1' and is not drawn.

In the main scenario the rule, not the road, limits throughput: road changes raise T1' while reservation barely moves



■ T1' (10 seeds) ■ Reservation, K = 60 ■ Reservation, K = 120

Three changes to road 161

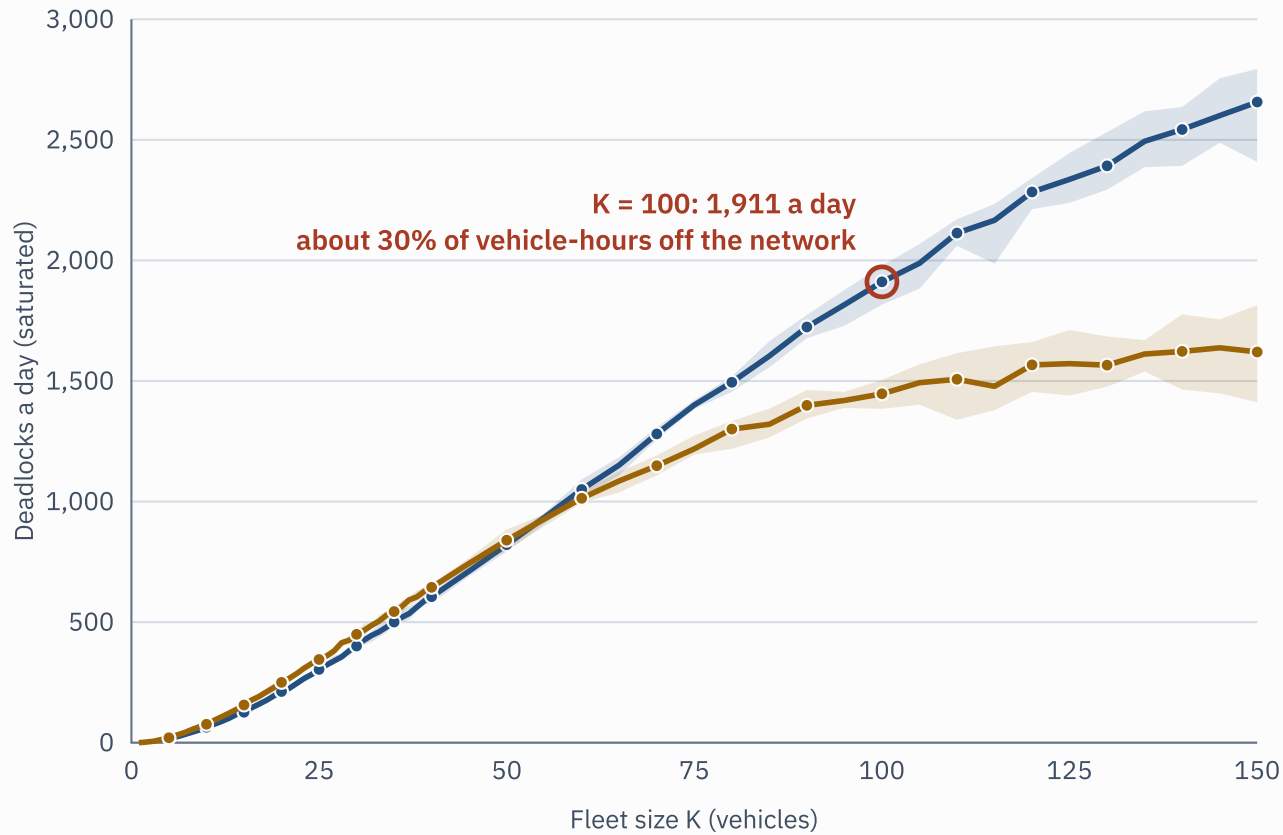
Widening to 12 m, segmented blocking at the stops, and two passing bays on top: T1' rises 5–8%, while saturated reservation throughput moves only +0.2% to +2.4%.

Small instances (checked by hand)

A 12 m single-lane corridor: reservation 19 a day, a hand-built feasible schedule 31.6, T1' 37.9. A crossroads: reservation 70, segment request (reference rule, teleport clearing) 105, T1' 187.

The cause is a rigid chain: reservation books empty run, loading, loaded run and unloading as one chain, so the booked road sits idle while the vehicle loads or unloads. With the plateau far below the bound, orchestration comes first.

Naive segment request deadlocks; it survives only by “teleport” clearing, so it is a reference rule



Okpo Yantai

26 / 83

Okpo deadlocks a day on a regular / peak day (97 / 194 tasks, the 4-vehicle subset of Yim 2008, for comparison only)

1,911 / 2,657

Deadlocks a day at saturation, K = 100 / 150

≈ 30%

Vehicle-hours teleported off the network at K = 100, at least (478 / 1,600)

The proof of T1' does not cover teleporting, and the manifesto bars the teleport rule from supporting any claim on its own. Hence physical rules: safe release, no clearing, physical clearing (T22). Support from ports: a terminal network deadlocks from 30 vehicles without keep-clear hatching at junctions (Zhou et al. 2017).

Sources: thesis-core 83c16bf, batches and conditions as on page 4. Segment request (reference rule, teleport clearing): at each deadlock one vehicle releases all its resources, waits 900 s on open ground and requests again from where it was (no published value for 900 s, T17). Deadlocks are means of 10 seeds, shading min-max; regular and peak are medians over the K grid. Vehicle-hours off the network = deadlocks × 900 s ÷ (K × 16 h), a lower bound. The author's guidance of 9 Oct 2026, §2.2.

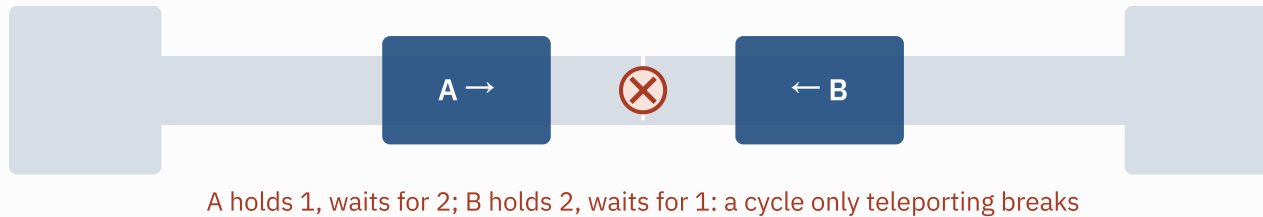
Rules in use and rules to build: from rigid reservation and a reference rule to provably safe orchestration

RULE	HOW IT RELEASES	DEADLOCK	LIMITATION	USE IN CHAPTER 4	STATUS
Whole-route reservation	Books the whole route before a loaded departure; leaves just in time	none	Rigid chain; plateau about 36% of T1'	Today's lower edge; baseline	done, T18
Segment request (reference rule, teleport clearing)	Requests one segment at a time; at a deadlock one vehicle leaves for 900 s	yes	Outside the T1' proof	Reference only	done, T17
Segment request, no clearing	As above, but deadlocked vehicles keep the road	yes	Throughput collapses once stuck	Measures naive deadlocks (H5, first criterion)	T22 running
Safe release, segment_safe	Grants a request only if the system stays in a safe state (Banker-type)	none	A safety check per request; heavy computation	Candidate “best deadlock-free rule”: the interval's lower edge and chapter 4's start	T22 running
Segmented blocking + direction locks	Long roads cut into blocks at their stops; in one lane, follow in the same direction, wait for the other	no head-on	Block length open (T25 scan)	The block-granularity lever; Okpo's main scenario becomes segmented	T22 running
T5: reserve to the next safe holding point	Before a loaded departure, books the road up to the next safe holding point and a slot there	none (to prove)	Holding-point placement and capacity set the loss	A safety shell around any heuristic; loss measured against T1'	planned

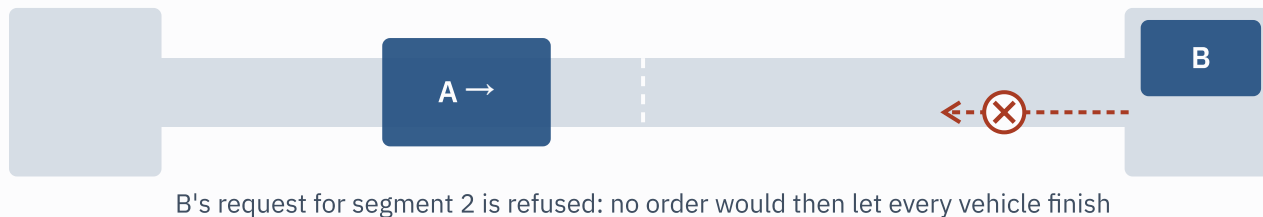
Sources: thesis-core 83c16bf (T17, T18 reports); T22 task note (project control runs/2026-10-09_T22_prompt.md §2; running, no results yet); the author's guidance of 9 Oct 2026, §2.2 (segmented blocking, direction locks) and the author's imagery check (neighbouring sites stay reachable while a vehicle works at the dock edge of road 161, so from T22 on Okpo's main scenario is segmented at the stops, with the whole segment as comparison); T5 wording per manifesto v1.

Two provably deadlock-free ways to release: a safe-state check, and reservation to the next safe holding point (T5)

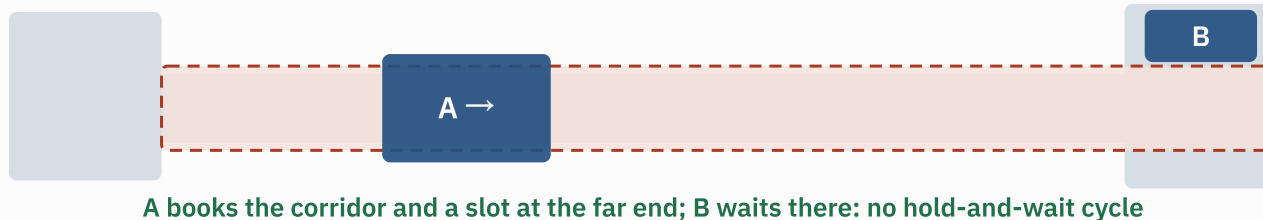
(a) Naive segment request



(b) Safe release, segment_safe: check the state first



(c) T5: book up to the next safe holding point

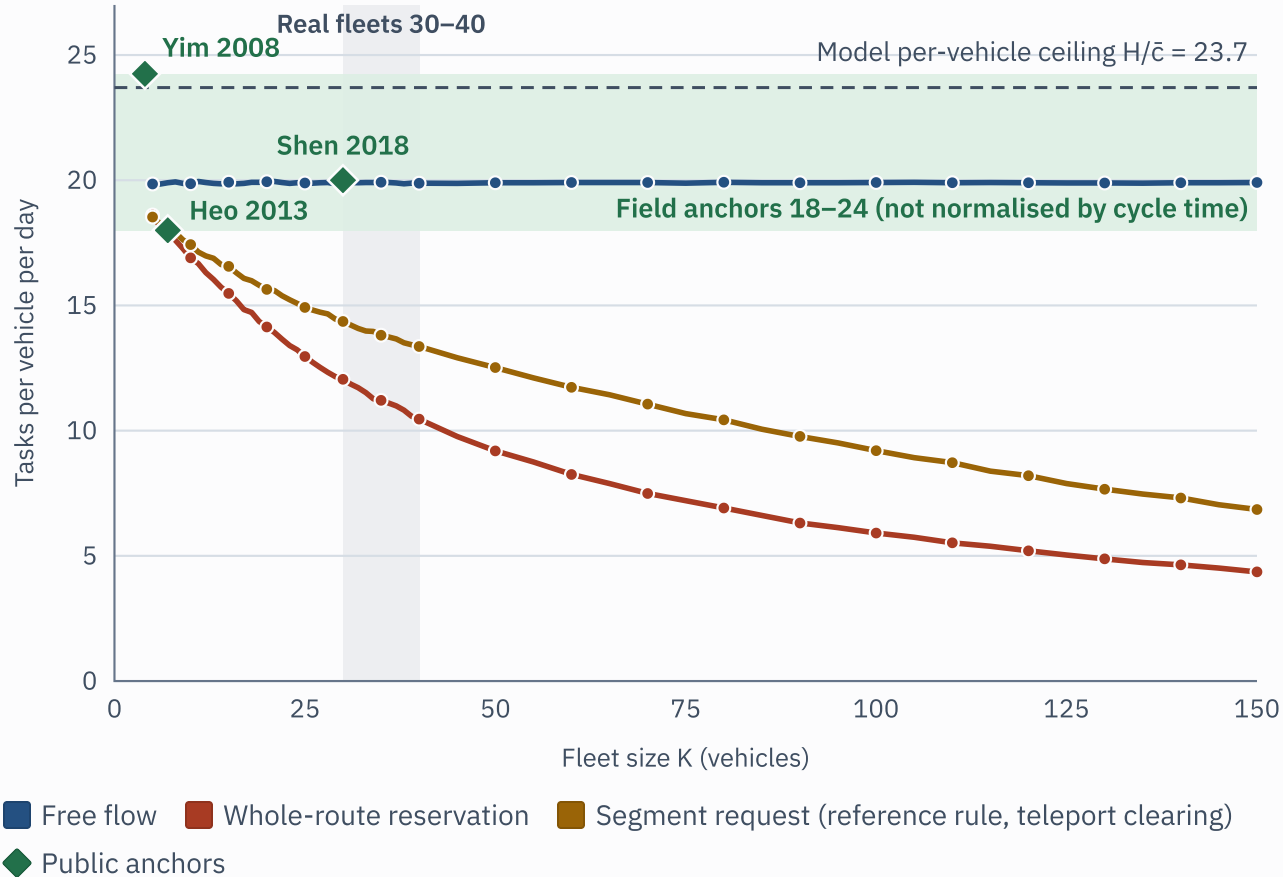


T5, safe orchestration: reserving to the next safe holding point never deadlocks, with a loss bound relative to the upper bound.

Proof route (plan)

- **No deadlock:** a loaded vehicle is always at a holding point or on road it has booked up to the next one, slot included, so no hold-and-wait cycle forms. Builds on the Banker's algorithm (Habermann 1969), resource allocation systems (Reveliotis 2000, 2005) and AGV reservation (Kim et al. 2006).
- **Loss bound:** T5 meets the conditions (A1)–(A3) of T1', so T1' bounds it; the loss is road held in advance, and max-plus gives a given order's throughput to compare with T1'.
- **Shipyard specifics:** holding points are off-road sites, dock-front bays or passing bays; cranes are released on a timetable, so the check treats them as available sooner or later.

The goal: push the lower edge towards field per-vehicle productivity; H5 asks what safety costs



12.1 → 18-24

Per vehicle per day on Okpo at K = 30: whole-route reservation 12.1, segment request (reference rule, teleport clearing) 14.4, free flow 19.9; field anchors 18-24

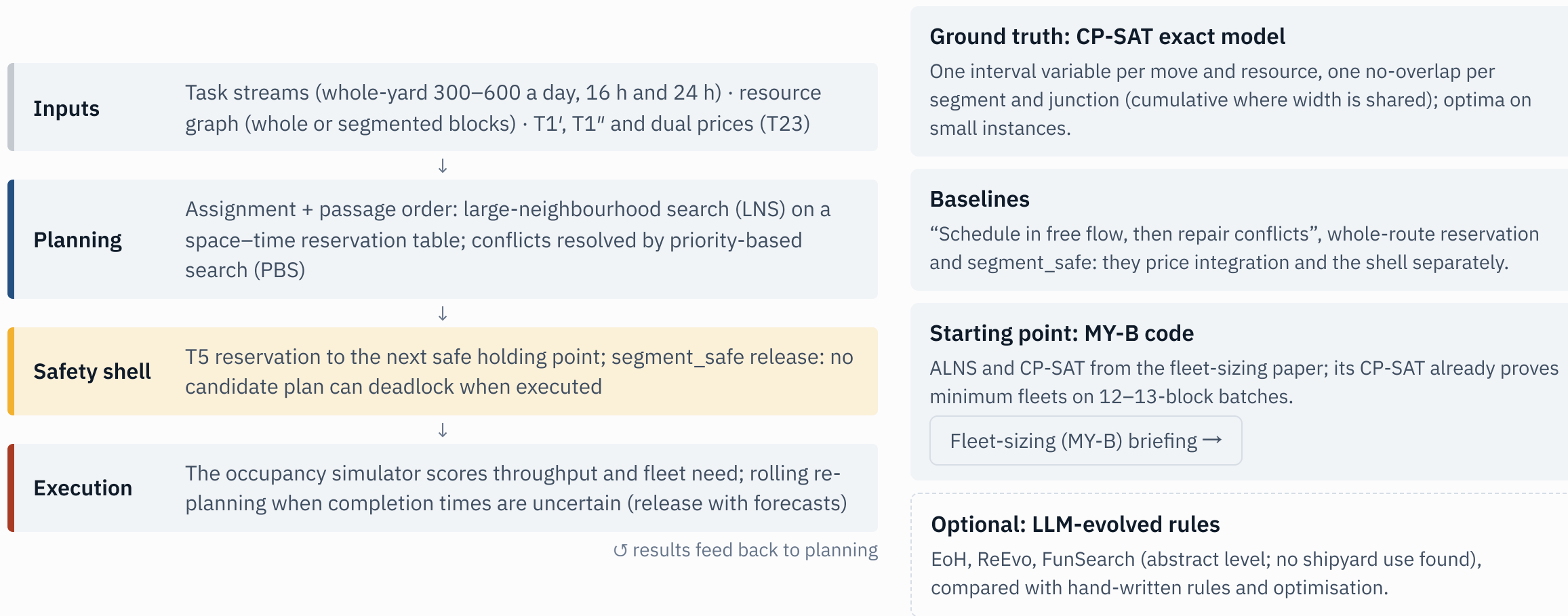
Compare after normalising by cycle time (distance, loading). If a gap remains, both readings go into the thesis: the model is conservative (whole-segment occupancy, empties occupying road, shifts, width mix), or the field orchestrates better (signallers, right of way, one-way loops). The latter is what chapter 4 aims to explain and reach.

H5 safe orchestration (manifesto v1)

Criteria: deadlocks are counted under no-teleport rules; the cost of safety is the best safe rule's loss relative to the network bound. **If it holds:** safe orchestration is cheap. **If not:** a formula for the cost of safety and its conditions.

Sources: thesis-core 83c16bf, Okpo main scenario, saturated (conditions as on page 4); per-vehicle ceiling $H/\bar{c} = 960 \text{ min} \div \text{mean cycle}$. Anchors: Yim 2008, 97 tasks / 4 vehicles (one day's subset at a large yard); Shen 2018, about 600 / about 30; Heo 2013, Subic 126 / 7 (2 auxiliary); none is Okpo itself, and T24 is checking the originals. "Real fleets of about 30-40": Shen 2018; 33-40 at Hyundai Heavy is press, to be checked. Goal from the author's guidance of 9 Oct 2026, §4.3, the two readings from §2.1; H5 per manifesto v1.

Methods: a safety shell around an integrated search over task assignment and passage order



Sources: thesis direction document (2nd ed.), §6 (algorithms) and §9 (MY-B code as chapter 4's start); the author's guidance of 9 Oct 2026, §4.3 (space–time reservation table, PBS, LNS); whole-yard task volumes from guidance §2.1 and T22; MY-B's CP-SAT scale from the fleet-sizing briefing (paper §4.2).

Every component has a mature source; what is new is an agent that widens when loaded, long lifts and a capacity bound

COMPONENT	TRANSFER SOURCE AND THEORY	WHAT IS NEW IN ROAD-OCCUPYING SHIPYARD TRANSPORT
Exact model	Job shop with blocking (Mascis & Pacciarelli 2002); OR-Tools CP-SAT	Width-shared resources; asymmetric loaded and empty occupancy; long holds at loading points
Integrated search	MAPF-LNS (Li et al. 2021); ALNS (Ropke & Pisinger 2006); MY-B code	Assignment and passage order searched together; empty runs need road too; fleet size is part of the decision
Conflict resolution	Priority-based search, PBS (Ma et al. 2019); rolling-horizon prioritised planning (Zheng et al. 2026)	Agents whose footprint multiplies when loaded; priorities tied to the reservation mode (whole route, to a holding point)
Safety shell	Banker's algorithm (Habermann 1969); resource allocation systems (Reveliotis 2000, 2005); AGV reservation with consistent priorities (Kim et al. 2006)	Holding points come from the network and vehicle class; the cost of safety is measured against the network bound T1'
Passing and blocking	Kiel Canal traffic with passing at sidings (Lübbecke et al. 2019, abstract level); railway block sections (UIC 406)	Block granularity as an orchestration variable, linked to T25's block-length scan
Optional: rule evolution	EoH, ReEvo, FunSearch (abstract level)	Readable release rules, tested for transfer across yards (links to chapter 6)

The closest work is in AGVs: Li et al. 2025 (preprint) schedule attachable AGVs deadlock-free with Petri nets and ALNS and draw fleet-sizing insights. We build on it for shipyard transport, where loads widen the footprint and losses are measured against the network bound.

Sources: direction document (2nd ed.), §6, §8.5, appendix; guidance of 9 Oct 2026, appendix B; library 04-023 (Li 2025) and 04-061 (Zheng 2026), full text. DOIs checked against Crossref. Abstract level: Mascis 2002, Reveliotis 2000 and 2005, Kim 2006, Lübbecke 2019, EoH, ReEvo, FunSearch; metadata only, to be read: Habermann 1969, Ma 2019, Li 2021.

Experiments: each one feeds a figure or table of chapter 4

#	EXPERIMENT	OUTPUT (CHAPTER 4)	NEEDS	STATUS
E1	The deadlock-free lower edge: segment_safe and reservation × whole / segmented blocks, both yards' saturated curves; deadlock counts with the no-clearing switch	Fig. 4-1 capacity interval; Table 4-1 deadlocks under no-teleport rules (H5, criterion 1)	T22	running
E2	Capacity orchestration can recover: a chain bound for reservation-type rules; max-plus throughput of T5 for a given order	Fig. 4-2 Okpo's gap split into rule loss and bound slack	T18 §7; T23	planned
E3	Small-instance ground truth: CP-SAT optima for a corridor, a crossroads, the three-resource dock network and small yard slices	Table 4-2 gaps between optimum, rules and heuristics	T18; T25	planned
E4	Integrated search (LNS + PBS + shell) against the three baselines; whole-yard volumes of 300 / 450 / 600 a day on both yards, 16 h and 24 h	Fig. 4-3 daily volume against fleet need; Table 4-3 cost of safety (H5, criterion 2)	T22; T23	planned
E5	Holding points and block granularity: holding-point spacing, holding capacity S, block length	Fig. 4-4 cost of safety against holding-point spacing and block length	T25; T26	planned
E6	Rolling online: release with forecasts under completion-time forecast errors	Fig. 4-5 throughput against forecast error	E4	planned
E7	External check: per-vehicle productivity normalised by cycle time against field anchors	Table 4-4 external check (same basis as chapter 3)	T24	planned

The first step is a constructive pilot: get CP-SAT and LNS running on E3's small instances, tune parameters and produce the first gap table. Small gaps: scale to both yards. Large gaps: revise the neighbourhoods and the shell, then continue.

Sources: the author's guidance of 9 Oct 2026, §§2.1–2.5, 4.1, 4.3; thesis-core 83c16bf docs/T18_report.md §§2.3, 7; task board tasks.md, T22–T26 (9 Oct 2026). Figure and table numbers are planned numbers.

What already holds, what is running, and risks with mitigations

Holds (chapter 3, 83c16bf, reviewed)

Interval readings; the rule limits the main scenario; the reference rule depends on teleporting; the rigid chain on small instances; T1' holds for any rule meeting (A1)–(A3), T5 included (loaded routes fixed)

Running (dispatched 9 Oct 2026)

T22: segment_safe, the no-clearing switch, segmented blocking with direction locks, whole-yard volumes. T23: system T1', route-free T1'', top-10 dual prices. T24: anchors and transfer sources checked

Planned

T25: block-length scan (capacity against granularity), the three-resource dock network. T26: finite holding area S and $K_0(S)$. Chapter 4 prototype: CP-SAT, LNS, the T5 proof

RISK	MITIGATION
T1' assumes fixed loaded routes; with detours in chapter 4 it is no longer a bound	Use route-free T1'' (T23) as the upper edge; T1' – T1'' prices route choice
segment_safe checks safety on every request, slow at K = 150 over 8 days	Incremental checks, only when a resource would fill; if needed, a cheaper sufficient condition, stating where it is more conservative
CP-SAT solves only small instances	Small instances give ground truth; at scale, LNS with its gap to T1' and T1'' reported
Field anchors are not Okpo's own; basis and shifts unclear	T24 checks the originals; normalise by cycle time; write both readings of any remaining gap
AGV deadlock-free scheduling exists (Li et al. 2025, preprint)	Cite it as a transfer source; what is new is load-dependent occupancy, the cost of safety against the network bound and cross-yard tests

Candidate journals: a few options, to be chosen when the paper is written

Preferred: theory and methods

- *Transportation Science*
T5 alongside exact and heuristic algorithms; the original ALNS paper (Ropke & Pisinger 2006) appeared here
- *Transportation Research Part C*
New methods for traffic operations and control; timetable-free railway capacity (Jensen et al. 2020) appeared here

Alternatives: OR and automation

- *IEEE Transactions on Automation Science and Engineering*
Deadlock avoidance and scheduling in automated material handling
- *European Journal of Operational Research*
Scheduling with blocking (Mascis & Pacciarelli 2002) appeared here
- *Computers & Operations Research*
Neighbourhood search and constraint programming (Pisinger & Ropke 2007) appeared here

Application-oriented

- *IEEE Transactions on Intelligent Transportation Systems*
Online orchestration and its implementation
- *International Journal of Production Research*
Shipyard production logistics; Roh & Cha 2011 on block transport scheduling appeared here

How to choose: theory and methods once T5 is proved with a loss bound; OR venues if algorithms and the two-yard experiments lead; application venues if the online system and field comparison lead. The direction document listed TR-C / T-ASE / EJOR, all included here.

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Other: UIC (2013) *UIC Code 406: Capacity*, 2nd ed.; the CP-SAT solver of Google OR-Tools; KED Global (2024) report on a Hanwha Ocean transporter and its signallers.

In short: make orchestration deadlock-free, then push the lower edge up

Grounds

The rule sets the interval's lower edge; the reservation plateau is about 36% of T1'; naive segment request avoids gridlock only by teleporting.

Approach

T5 and safe release as the shell; integrated search over assignment and passage order inside; CP-SAT ground truth; rolling online.

Outputs

The interval with a deadlock-free lower edge, H5's cost of safety, a proof of T5, a chapter 4 paper.

[Back to the thesis briefing →](#)[Capacity interval →](#)[Dock mouth →](#)[Fleet-network co-design →](#)[Fleet-sizing \(MY-B\) briefing →](#)

Where the network binds, schedule the road rather than the vehicles: orchestration decides how much of the bound is achieved.

Thank you.